Optimal Field Sampling Framework for Disease Surveillance of Clonally Propagated Plants

DisameR package

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Why This Problem Matters

- Epidemics in biological systems — human, animal, and plant — emerge from the interaction of **local transmission mechanisms** and **large-scale spatial structure**.
- Classical models explain **how diseases spread**, but rarely address:
  - **What we fail to observe**
  - **Where to look next**
  - **How sampling changes decisions**
- In real surveillance systems, data is:
  - Sparse
  - Delayed
  - Biased toward visible symptoms

Core Question: How can dynamical systems, statistical inference, and decision theory be unified into a *single mathematical framework for adaptive epidemic surveillance*?

The Mathematical Gap

Three Mathematical Layers — Rarely Coupled in a Single Framework

Dynamical Systems

- Delay differential equations
- Spatial force of infection
- Latent and symptomatic structure
- Well-posedness, invariance, $\mathcal{R}_0$

Statistical Inference

- Partial observability
- Posterior on latent burden θ_j
- Sampling likelihood
- Uncertainty near thresholds

Decision Theory

- Action space $\{A_0, A_1, A_2\}$
- Transmission-based thresholds
- Bayesian loss function
- EVSI and optimal sample size

Central Challenge: Can we construct a *unified mathematical framework* in which delay-driven epidemic dynamics, Bayesian inference on hidden states, and optimal surveillance decisions are solved as a single, spatially coupled control problem?

Epidemics as Partially Observed Dynamical Systems

Key Structural Feature: Hidden States

- In many biological epidemics, only a **fraction of infections** are observable due to: i) Asymptomatic phase ii) Delayed symptom expression iii) Incomplete surveillance
- This gives rise to a **latent dynamical system** described by:

$$\begin{aligned} \text{State dynamics:} \quad & X_{t+1} = f(X_t, \theta, \xi_t) \\ \text{Observation model:} \quad & Y_t = g(X_t, \varepsilon_t) \end{aligned}$$

Where:

- X_t: latent (unobserved) system state at time t
- Y_t: observed data at time t
- θ: vector of fixed model parameters
- ξ_t: process noise (intrinsic/environmental stochasticity)
- ε_t: observation noise (measurement/sampling error)

Implication: Inference, prediction, and control are fundamentally limited by the *unobserved* components of the system.

Spatial Structure and Multiscale Transmission

Two Coupled Transmission Regimes:

- **Local dynamics:**
  - Vector-mediated spread
  - Short-range spatial coupling
  - Continuous-time processes
- **Long-range dynamics:**
  - Movement of hosts or planting material
  - Network-driven transmission
  - Discrete stochastic jumps

Mathematical Challenge: How do we couple **reaction–diffusion dynamics** with **network-driven jump processes** in a statistically identifiable way?

Why Classical Sampling Frameworks Fail

Traditional surveillance assumes:

- observable infection reflects true infection,
- prevalence estimation is the primary goal,
- farms are sampled independently,
- fixed sample sizes are sufficient.

But in clonal epidemics:

- latent infections dominate transmission,
- symptom visibility is delayed,
- spatial spillover creates network dependence,
- sampling must change management decisions.

Core Computational Problem

How should (m, n_j) be allocated under hidden spatial transmission?

Joint Optimization of Farms (m) and Plants (n_j)

Two coupled surveillance decisions are optimized simultaneously:

- m: number of farms to visit
- n_j: number of plants sampled within farm j

Optimization objective:

$$\max_{\{n_j\}} \sum_{j \in \mathcal{J}} \left(\mathrm{EVSI}_j(n_j) - C(n_j) \right)$$

subject to:

$$\sum_{j \in \mathcal{J}} \mathbb{I}(n_j > 0)(c_{\text{farm}} + n_j c_{\text{plant}}) \leq B$$

Key Innovation:

Sampling effort emerges from:

- latent burden,
- spatial coupling,
- decision thresholds,
- and uncertainty propagation.

Result: Optimal surveillance becomes adaptive, spatially targeted, and decision-aware.

R Package Computational Architecture

Core modules of the package

Epidemic Engine

- Spatial DDE solver
- Metapopulation dynamics
- Latent state propagation

Inference Layer

- Hierarchical likelihood
- Posterior updating
- Latent burden estimation

Decision Engine

- EVSI computation
- Bayesian loss minimization
- Threshold-based actions

Optimization Layer

- Farm allocation (m)
- Plant allocation (n_j)
- Budget-constrained optimization

Dynamics $\rightarrow$ Inference $\rightarrow$ Decision $\rightarrow$ Optimization

Package Workflow

User workflow implemented in the R package

1. Import farm coordinates and epidemiological parameters
2. Simulate latent epidemic dynamics
3. Compute posterior infection burden
4. Evaluate decision thresholds
5. Optimize:

$$(m, n_j)$$

   under budget constraints
6. Generate adaptive surveillance maps

Outputs

- Optimal farms to visit
- Optimal plants per farm
- EVSI surfaces
- Spatial risk maps
- Decision recommendations

From the Original Spatial DDE Script to the JTB Pipeline

Old Framework

- Spatial DDE epidemic model only
- Seasonal simulation loop
- Fixed surveillance interpretation
- Static prevalence visualization
- Thresholds computed separately
- Sampling optimization disconnected from epidemic dynamics
- No explicit budget constraints
- No adaptive surveillance policy
- No within-farm dependence correction
- Independent analyses:
  - DDE dynamics
  - R_0 analysis
  - sensitivity analysis
  - spatial visualization

Updated Pipeline

- Fully integrated bio-economic framework
- Biological engine + decision engine
- Automatic threshold extraction:

$$\theta_1, \theta_2$$

- Responsive optimization under budget constraints
- Dynamic surveillance allocation:

$$(n^*, m^*)$$

- Explicit value-of-information framework
- Regional detection probability:

$$P_{\text{det,total}}$$

New Functions Introduced in the Updated Pipeline

Function	Role in the Framework
`cutting_model()`	Core spatial DDE biological engine with latency and spatial coupling
`get_thresholds()`	Automatically computes epidemiological decision thresholds θ_1 and θ_2
`find_optimal_policy()`	Main constrained optimizer returning optimal surveillance allocation (n^*, m^*) under budget constraints
`history()`	Initial history generator for delayed dynamics
Sensitivity pipeline	Stress-testing of optimal policy across parameter uncertainty
Budget-response engine	Evaluates surveillance scaling across operational budgets
Visualization pipeline	Produces integrated policy frontiers, latency traps, and spatial information maps

Functions to Integrate into the Future R Package

Suggested Package Structure

1. Biological Dynamics Module
   - `cutting_model()`
   - `history()`
   - `simulate_dde()`
   - `compute_prevalence()`
2. Threshold & Epidemiological Metrics
   - `get_thresholds()`
   - `compute_R0()`
   - `detectability_index()`
3. Surveillance Optimization Engine
   - `find_optimal_policy()`
   - `compute_ENGS()`

Vision for the Future Package

Proposed Objective

Develop an integrated R package for:

- spatial epidemic simulation,
- surveillance optimization,
- value-of-information analysis,
- and adaptive bio-economic decision support.

Potential Package Workflow

$$\texttt{simulate_dde()} \rightarrow \texttt{get_thresholds()} \rightarrow \texttt{find_optimal_policy()} \rightarrow \texttt{plot_results()}$$

Key Innovation

The package would bridge:

$$\textbf{epidemiology} + \textbf{economics} + \textbf{surveillance design}$$

within a single operational framework.